

Résumé Bastian Kuth

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*26.05.1998 in Bamberg, Germany

Work Experience

- January 2023 – March 2026
Research Associate *full-time*
Coburg University of Applied Sciences
Supervisor: Prof. Dr.-Ing. Quirin Meyer    
Project GeoFlow – Data-centered real-time geometry processing on graphics cards
Worked closely with AMD team to research novel techniques for real-time geometry processing using Direct3D12 GPU work graphs and mesh shaders.
Delivered prototypes, benchmarks, tutorials, demos (GDC), courses (GPC, SIGGRAPH), articles (GPUOpen), and four award-winning papers (HPG, VMV, C&G) featuring these technologies.
- March 2024 – August 2024
Contract Lecturer 2 SWS
Coburg University of Applied Sciences
3D Gaussian Splatting in Direct3D12
Supervised visual computing students developing a Gaussian splatting renderer in Direct3D12 and gave lectures about real-time rendering.
- March 2022 – December 2022
Research Assistant *part-time*
Nuremberg Institute of Technology: [Game Tech Lab](#)
Supervisor: Prof. Dr. rer. nat. Bartosz von Rymon Lipiński    
Project IBMPP – Image-Based Media Production Pipeline
Conceptualized and implemented a VR renderer and genetic optimization pipeline for densely feature-matched images (OpenGL 4.5, OpenXR, C++). Co-authored two papers.
- February 2020
Research Assistant *part-time*
Coburg University: [mixedrealitylab](#)
Supervisor: Prof. Dr. techn. Jens Grubert    
Project Virtual Office
Developed VR prototypes (Unity, C#), assisted in VR user studies, co-authored a paper.
- October 2019 – February 2020
Internship Semester *full-time*
Siemens Healthineers – Product Development
Supervisor: Dr.-Ing. Philip Mewes   
Project [AI-Rad Companion](#)
Responsible for technical setups of user studies required for FDA approval of medical products. Setup included a simulated hospital environment with PACS, [teamplay](#), and AI Rad Companion platform to be used by Radiologists in these studies.
- January 2019 – March 2019
Research Assistant *part-time*
Coburg University: [mixedrealitylab](#)
Supervisor: Prof. Dr. techn. Jens Grubert    
Project Virtual Office
Developed VR prototypes (Unity, C#), installed an OptiTrack MoCap system, co-authored a paper.
- Summer 2018 – Fall 2018
Working Student *part-time*
Siemens Healthineers – Product Innovation
Supervisor: Dr.-Ing. Peter Fischer   
Project Surgery Robotics
Implemented an evaluation pipeline for x-ray marker-based registration (Python, OpenCV). Generated synthetic X-ray data (Blender).
- Summer 2017 – Fall 2017
Vacation Worker *full-time*
Siemens Healthineers – Product Innovation
Supervisor: Dr.-Ing. Peter Fischer   
Project Surgery Robotics
Conducted experiments on the feasibility of 3D printed X-ray markers for surgical robotics (C++, OpenCV, ViSP). Was entrusted with independent operation of an ARTIS pheno X-ray system.

Education

- 2023 – Present **Doktor-Ingenieur (PhD)**
Coburg University of Applied Sciences • University of Erlangen–Nuremberg (FAU)
Thesis: Real-Time Geometry Amplification on Graphics Hardware
Advisors: Prof. Dr.-Ing. Marc Stamminger   • Prof. Dr.-Ing. Quirin Meyer   
- 2021 – 2022 **Master of Science in Computer Science**
Nuremberg Institute of Technology Georg Simon Ohm
Thesis: Real-Time Image-Based 3D-VR-Rendering of Open Environments
Advisor: Prof. Dr. rer. nat. Bartosz von Rymon Lipiński   
Thesis Grade: 1.0 Overall Grade: 1.0 (best of the year)
- 2017 – 2021 **Bachelor of Science in Computer Science**
Coburg University of Applied Sciences
Thesis: Compression of Vertex Blending Attributes
Advisor: Prof. Dr.-Ing. Quirin Meyer   
Thesis Grade: 1.0 Overall Grade: 1.3

Skills

Programming

Proficient in: C++, C, Python, GLSL, HLSL

Familiar with: Java, C#, Haskell, JavaScript

Tools, Libraries and Software

Blender, Git, Gimp, L^AT_EX, Overleaf, Gurobi, SCIPopt, Direct3D12, OpenGL, OpenCV, OpenXR, Vulkan, CUDA, CMake, Unity, NumPy, SymPy, ImGui, Real Virtuality Engine, Matplotlib, Tensorflow, Keras, Pytorch, Adobe Premiere, Visual Studio, VS Code

Languages

German (native), English (fluent)

Awards

- June 2025 **Wolfgang Straßer Best Paper Award (2nd place)**
At [High-Performance Graphics 2025](#)
For *Real-Time GPU Tree Generation*
- September 2024 **Best Paper Award**
At [Vision, Modeling, and Visualization 2024](#)
For *Towards Practical Meshlet Compression*
- July 2024 **Wolfgang Straßer Best Paper Award (1st place)**
At [High-Performance Graphics 2024](#)
For *Real-Time Procedural Generation with GPU Work Graphs*
- June 2023 **Best Computer Science Master's Degree of the Year**
At [Nuremberg Institute of Technology Georg Simon Ohm](#)
- June 2023 **Wolfgang Straßer Best Paper Award (3rd place)**
At [High-Performance Graphics 2023](#)
For *Edge-Friend: Fast and Deterministic Catmull-Clark Subdivision Surfaces*
- June 2023 **DATEV Sponsorship Award**
At [Nuremberg Institute of Technology Georg Simon Ohm](#)
For Master's Thesis *Real-Time Image-Based 3D-VR-Rendering of Open Environments*
- May 2022 **Best Paper Award**
At [ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games 2022](#)
For *Permutation Coding for Vertex-Blend Attribute Compression*
- July 2021 **Wolfgang Straßer Best Paper Award (1st place)**
At [High-Performance Graphics 2021](#)
For *Vertex-Blend Attribute Compression*
- October 2019 **Best Paper Award**
At [International Symposium on Mixed and Augmented Reality 2019](#) (IEEE ISMAR)
For *ReconVigURation: Reconfiguring Physical Keyboards in Virtual Reality*

Public Talks

- 2025 **GPU Work Graphs**
Course at SIGGRAPH 2025
- 2025 **Real-Time GPU Tree Generation**
Paper presentation at HPG 2025 in Copenhagen, Denmark
- 2024 **GPU Work Graphs - mastering the future of GPU programming**
Masterclass at GPC 2024 in Breda, Netherlands
- 2024 **Towards Practical Meshlet Compression**
Paper presentation at VMV 2024 in Garching, Germany
- 2024 **Real-Time Procedural Generation with GPU Work Graphs**
Paper presentation at HPG 2024 in Denver, USA, [recording](#)
- 2023 **Edge-Friend: Fast and Deterministic Catmull-Clark Subdivision Surfaces**
Paper presentation at HPG 2023 in Delft, Netherlands, [recording](#)

Publications

- 2025 **Real-Time GPU Tree Generation**
Bastian Kuth, Max Oberberger, Carsten Faber, Pirmin Pfeifer, Seyedmasih Tabaei, Dominik Baumeister, Quirin Meyer *High-Performance Graphics - Symposium Papers (HPG)*
- 2025 **EGO-VC – Evolutionary GPU-Optimization of Visual Correspondences for Image Alignment**
Thomas Chang, Karl Hartmann, [Bastian Kuth](#), Simon Seibt, Bartosz von Rymon Lipiński *WSCG 2025* (follows section 3 of my master thesis)
- 2025 **Multidimensional Image Morphing-Fast Image-based Rendering of Open 3D and VR Environments**
Simon Seibt, [Bastian Kuth](#), Bartosz von Rymon Lipiński, Thomas Chang, Marc Erich Latoschik *Virtual Reality and Intelligent Hardware (VRIH)* (follows section 2 of my master thesis)
- 2024 **Towards Practical Meshlet Compression**
[Bastian Kuth](#), Max Oberberger, Felix Kawala, Sander Reitter, Sebastian Michel, Matthäus Chajdas, Quirin Meyer
Vision, Modeling, and Visualization (VMV)
- 2024 **Real-Time Procedural Generation with GPU Work Graphs**
[Bastian Kuth](#), Max Oberberger, Carsten Faber, Dominik Baumeister, Matthäus Chajdas, Quirin Meyer
Proceedings of ACM on Computer Graphics and Interactive Techniques (PACMCGIT / HPG)
- 2023 – 2024 **Mesh shaders on RDNA™ graphics cards**
Max Oberberger, [Bastian Kuth](#), Quirin Meyer
[GPUopen](#) (not peer-reviewed)
- 2023 **Edge-Friend: Fast and Deterministic Catmull-Clark Subdivision Surfaces**
[Bastian Kuth](#), Max Oberberger, Matthäus Chajdas, Quirin Meyer
Computer Graphics forum (CGF / HPG)
- 2022 **Permutation Coding for Vertex-Blend Attribute Compression**
Christoph Peters, [Bastian Kuth](#), Quirin Meyer
ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games (I3D)
- 2021 **Vertex-Blend Attribute Compression**
[Bastian Kuth](#), Quirin Meyer
High-Performance Graphics - Symposium Papers (HPG)
- 2020 **Breaking the Screen: Interaction Across Touchscreen Boundaries in Virtual Reality for Mobile Knowledge Workers**
Verena Biener, Daniel Schneider, Travis Gesslein, Alexander Otte, [Bastian Kuth](#), Per Ola Kristensson, Eyal Ofek, Michel Pahud, Jens Grubert
IEEE Transactions of Visualization and Computer Graphics (TVCG)
- 2019 **ReconVigURation: Reconfiguring Physical Keyboards in Virtual Reality**
Daniel Schneider, Alexander Otte, Travis Gesslein, Philipp Gagel, [Bastian Kuth](#), Mohamad Shahm Damlakhi, Oliver Dietz, Eyal Ofek, Michel Pahud, Per Ola Kristensson, Jörg Müller, Jens Grubert
IEEE Transactions of Visualization and Computer Graphics (TVCG)

Voluntary Work

- 2025 **Paper Reviewer** 1 review at [Eurographics 2026](#)
- 2025 **Paper Reviewer** 3 reviews at [High-Performance Graphics 2025](#)
- 2025 **Paper Reviewer** 1 review at [Eurographics 2025](#)
- 2024 **Paper Reviewer** 1 review at [SIGGRAPH Asia 2024](#)
- 2024 **Paper Reviewer** 2 reviews at [High-Performance Graphics 2024](#)
- Fall 2016 – **Bundesfreiwilligendienst** *full-time*
- Spring 2017 Federal Agency for Technical Relief (THW) • Geschäftsstelle Bamberg

Video Game Modding

- 2019–2021 **Nassau 1715**
Our international team created a total conversion mod that transforms Arma 3 from a modern military simulation on a Greek island into one about pirates in the Bahamas. I handled most of the programming. Key features include a physically accurate sailing model and a custom interaction system. The mod has ~17k current subscribers or ~62k total downloads on the [Steam Workshop](#).
- 2017 **WMO – Walkable Moving Objects**
A mod that extends Arma 3's Real Virtuality Engine to support players riding moving objects, such as cars, boats, or airplanes. With the vanilla game engine, players just slide off or die when trying to board a vehicle. This mod fixes this. It currently has ~263k+ subscribers and ~538k+ total downloads on the [Steam Workshop](#).
- 2014–2019 **Loewenherz Altis Life**
Volunteered as developer, admin, and moderator for one of Germany's largest Altis Life servers — a 100+ player (MMO)RPG game mode for Arma 3. Contributed to scripting, texturing, 3D modeling, as well as gameplay/sound/level design. At its peak, the server database had 70,000 unique players and the community forum over 7,000 members.

Notable Media Coverage

- 2025 **TechPowerUp** [link](#)
Researchers Unveil Real-Time GPU-Only Pipeline for Fully Procedural Trees
- 2025 **PC Games Hardware** [link](#)
Von 35 GiB auf 51 KiB: Forscher demonstriert "Work Graphs" für 3D-Rendering von Bäumen
- 2025 **PC Gamer** [link](#)
Graphics researchers have created a GPU-run procedural algorithm for creating an equivalent 35.6 GB worth of trees, leaves, and bushes from just 52 kB of data
- 2025 **Tom's Hardware** [link](#)
AMD researchers reduce graphics card VRAM capacity of 3D-rendered trees from 38GB to just 52 KB with work graphs and mesh nodes — shifting CPU work to the GPU yields tremendous results
- 2024 **Heise Online** [link](#)
New GPU technology creates dynamic 3D worlds
- 2024 **Tom's Hardware** [link](#)
AMD shows off DX12-related rendering advances that make game engines more efficient and less dependent on the CPU — demo shows a 64% improvement